
Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix

Source: Our World in Data – Renewable Energy (Ember 2026; Energy Institute 2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global greenhouse gas emissions come
from burning fossil fuels for energy

5M+

premature deaths per year from fossil fuel air
pollution

Since the Industrial Revolution, most countries' energy systems have become dominated by fossil fuels.

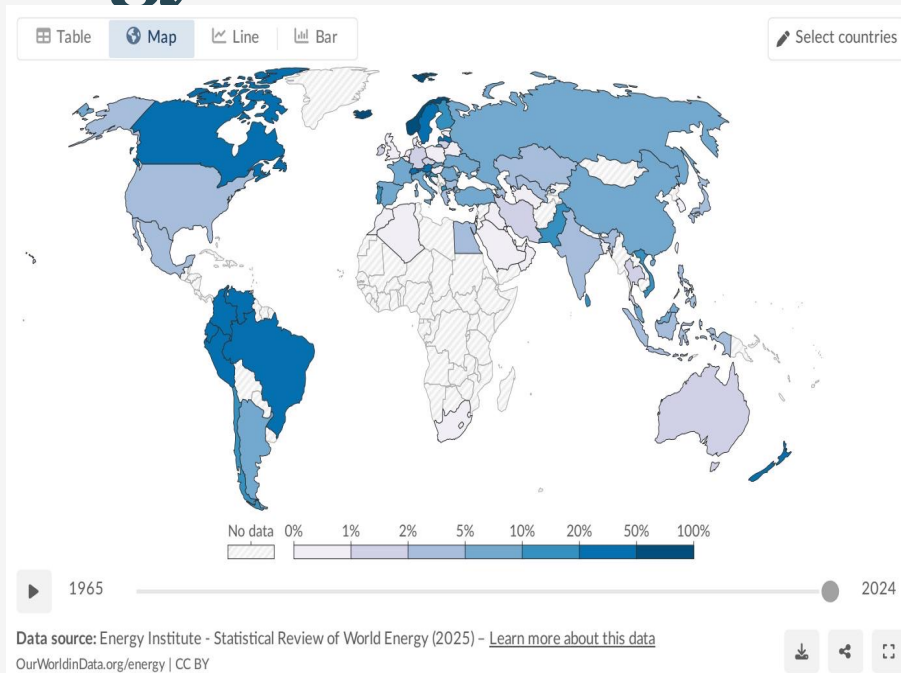
- Fossil fuel combustion drives climate change and local air pollution
- The world must rapidly shift to low-carbon sources — nuclear and renewables
- Renewables include solar, wind, hydropower, geothermal, bioenergy, wave, and tidal
- Traditional biomass is typically excluded from modern renewable statistics

Renewables Now Supply ~14% of Global Primary Energy

Approximately one-seventh of the world's primary energy now comes from renewable technologies (2024 data, substitution method).

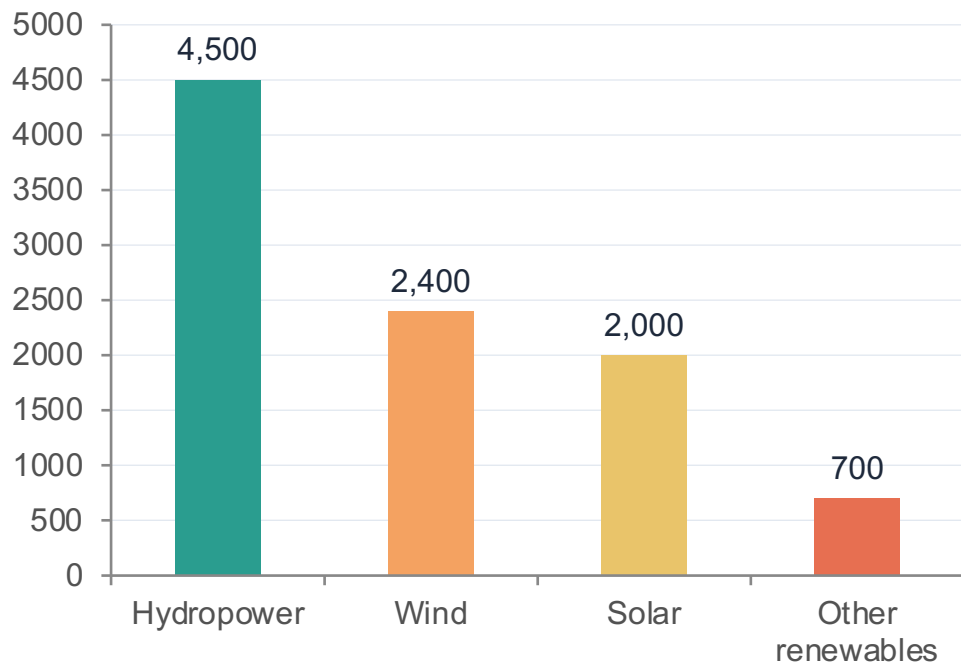
Primary energy encompasses electricity, transport, and heating combined. Because transport and heating are harder to decarbonize, the renewable share in total energy is lower than in electricity alone.

Regional variation is significant: Norway, Brazil, and New Zealand derive large shares from renewables, while fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



Technology	Generation (TWh)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other renewables	~700	~7%

Wind and solar combined now nearly match hydropower's output and are growing much faster. Solar in particular has seen dramatic growth since 2010, driven by falling costs and policy support.

Almost One-Third of Global Electricity Now Comes from Renewables

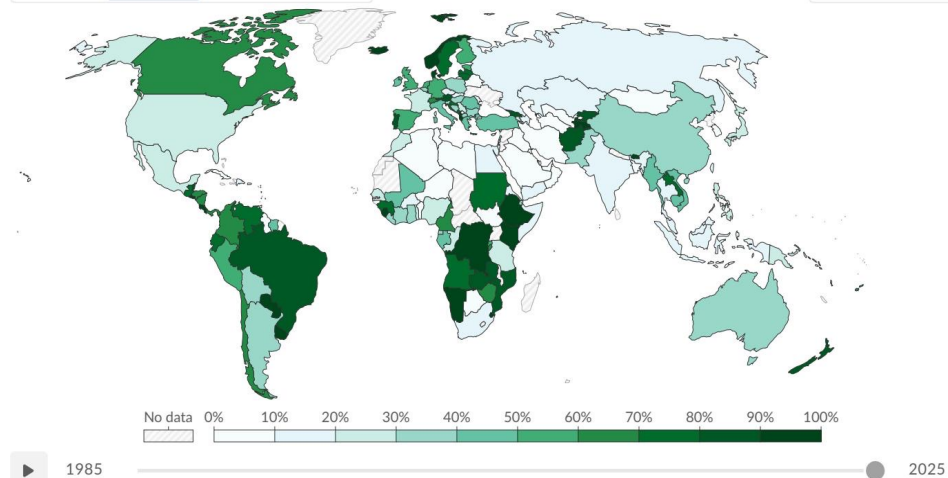
Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

Table Map Line Bar

Select countries



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Share of electricity from renewables by country, 2025

Country / Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Several African and Middle Eastern countries have shares below 10%, reflecting fossil fuel dependence and limited renewable investment.

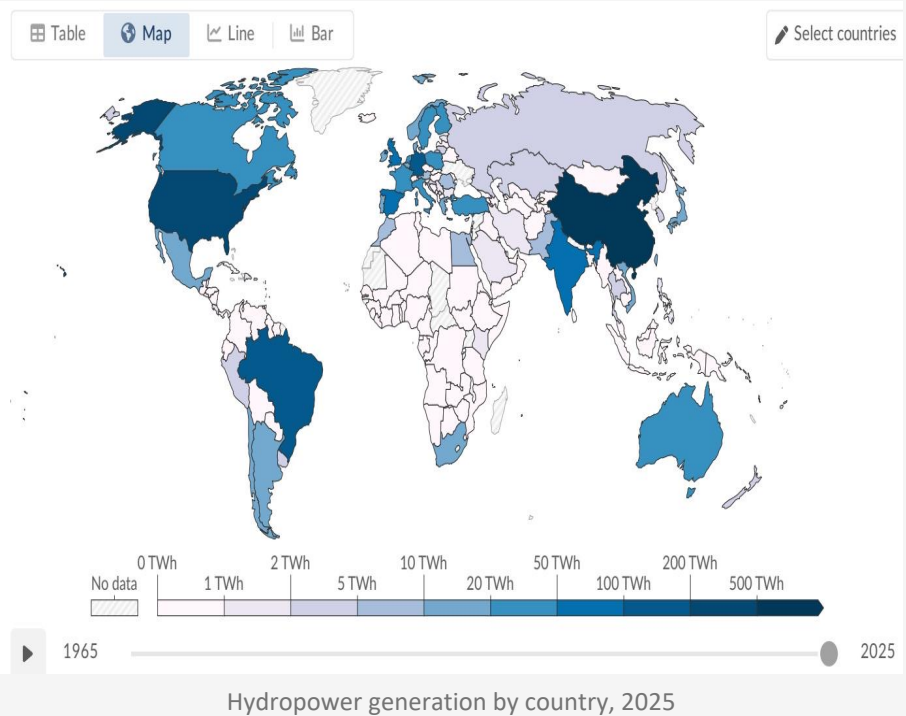
Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower remains the single largest renewable electricity source, accounting for roughly half of all renewable generation. It has been generating electricity at scale for over a century.

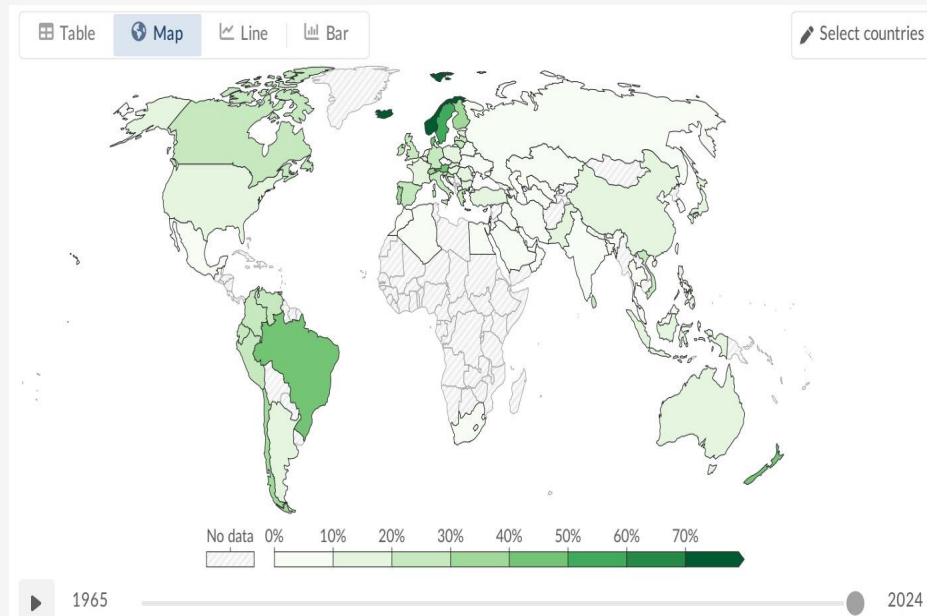
Top Producers

- China – largest global hydropower generator
- Brazil – abundant hydro resources
- Canada – significant hydroelectric capacity

~4,500 TWh generated globally | ~47% of all renewable electricity



Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country, 2025

Wind

~2,400 TWh

~25% of renewable electricity | Major contributor in Europe, China, and the United States (onshore and offshore)

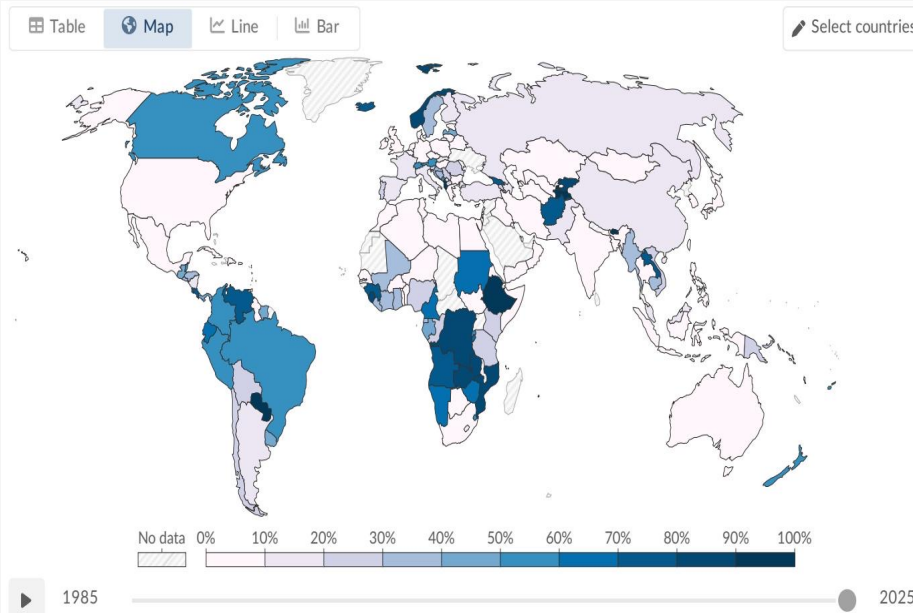
Solar

~2,000 TWh

~21% of renewable electricity | Fastest-growing source; dramatic growth since 2010 driven by falling costs and policy support

Combined, wind + solar now nearly match hydropower output

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Share of electricity from hydropower by country, 2025

Leaders

- Norway ~98%
- Brazil ~85%
- Canada ~65%
- Germany ~55%
- Denmark & Germany invested heavily in wind and solar

Lagging Regions

- Saudi Arabia <2%
- South Africa ~10%
- Much of Sub-Saharan Africa has minimal renewable electricity infrastructure
- Many Middle Eastern nations remain almost entirely fossil-fuel dependent

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14%

of global primary energy

Transport and heating lag far behind electricity, keeping the overall share low.

~30%

of global electricity

Electricity is where renewables have made the most progress — but it's not yet enough for climate targets.

Wind + Solar

fastest-growing technologies

Combined, they now produce nearly as much as hydropower and are accelerating.

Stark gaps

between regions persist

Scandinavia and Latin America exceed 60–90%; much of the Middle East and Africa remain below 10%.

The pace of deployment must accelerate dramatically to meet global climate and air quality goals.